

Braden McDorman

(405) 795-1800 | braden@mcdorman.io

An interactive version of this resume with notable projects, additional roles, and expanded role details is available at mcdorman.io/resume.

Summary

With **over 10 years of experience as a leader and individual contributor** in both startups and large organizations, I am a highly ambitious and adaptable engineer **capable of driving projects from zero to one**, including cross-functional alignment, design, planning, implementation, and delivery.

In my recent role I **spearheaded development and adoption of AI/ML for both engineering and customer teams, including agentic workflows and in-house ML models.**

Experience

Zoox – Engineering Manager – Foster City, CA ————— **Sep 2024 to Present**

Managing a six-person team of software and ML engineers building the simulation tools enabling millions of daily AV simulations, used by hundreds of engineers across dozens of teams to validate the vehicle's safety, service tooling, and collect performance metrics.

Shifted the team's technical focus from web development to generative AI: hired ML engineers, shipped agentic tooling that automated manual simulation workflows, and deployed in-house generative models to production for exploratory vehicle testing.

Zoox – Senior Software Engineer – Foster City, CA ————— **Jun 2023 to Sep 2024**

Designed and built a 3D visual editor for authoring simulation scenarios, used to validate vehicle safety and test autonomous vehicle components across Zoox's engineering teams.

Semio – Co-Founder & CTO – Los Angeles, CA ————— **Jan 2015 to Apr 2023**

Co-founded Semio, a social robotics software startup, growing it to \$1M in annual revenue, entirely bootstrapped. Created the OS, app ecosystem, and developer tools for social robots.

Architected and built all aspects of Semio's products: web front-end, back-end, cloud infrastructure, 3D simulation, social behavior models, and on-robot software. Managed contractors and interns. Aligned and iterated with Fortune 100 customer teams to address their needs.

KIPR – Lead Software Engineer – Norman, OK ————— **Jan 2015 to Jan 2016**

Led the design, development, and testing of the Wallaby robot controller for the Botball educational robotics program. This controller targeted novice K-12 and higher education students and is in use internationally by tens of thousands of students and educators.

Disney Research – Associate – Pittsburgh, PA ————— **May 2015 to Aug 2015**

Implemented and conducted studies on group formation recognition and planning algorithms for social robots in human spaces. Published in the Proceedings of the Human Robot Interaction 2017 conference.

Earlier Experience

Microsoft · Software Development Engineer Intern · Redmond, WA ————— May 2014 to Aug 2014

NewSpin 360 · Research and Development Software Engineer · Remote ————— Jan 2014 to Apr 2014

OU · Teaching Assistant · Norman, OK ————— Aug 2013 to Mar 2014

KIPR · Software Engineer · Norman, OK ————— Jun 2011 to Feb 2014

OU · Research Assistant · Norman, OK ————— Aug 2013 to Jan 2014
Plasma @ UMass · Google Summer of Code Student · Remote ————— Jun 2013 to Sep 2013
OU Health Sciences Center · Student Researcher · Oklahoma City, OK ————— Aug 2011 to May 2012

Advisory Experience

KISS Institute for Practical Robotics (KIPR) – Board Member – Remote ————— Apr 2021 to Present

Helping guide the mission and executive direction of KIPR, a 501(c)(3) nonprofit that runs the international Botball Educational Robotics Program, Junior Botball Challenge, and other STEM programs for K-12 educators and students, with particular focus on underserved communities.

KIPR – Technical Advisory Committee Chair – Remote ————— May 2019 to Present

Bobble Sports – Advisor – Remote ————— Jan 2020 to Apr 2022

Advising company founders on Bobble Sports' web-based event interaction system, including web frontend, app development, cloud backend, and high performance audio synthesis system.

Education

University of Southern California – Ph.D. in Computer Science - Incomplete — Aug 2015 to Dec 2016

Received the National Science Foundation Graduate Research Fellowship and pursued a Ph.D. in Computer Science with focus in social robotics. Advised by [Prof. Maja Matarić](#). Left the Ph.D. program *without receiving a degree* to pursue Semio, a robotics startup I co-founded, full-time.

University of Oklahoma – B.S. in Computer Science ————— Aug 2012 to May 2015

Received B.S. in Computer Science with minor in Mathematics. Instructor for the Computer Science Interview Preparation club. Selected participant in [DARPA Virtual Robotics Challenge](#).

Skills

Over 10 years

Programming Languages C | C++ | JavaScript | HTML

Platforms Linux | macOS

Tools Git | CMake | GSuite | Microsoft Office

Competencies Technical Leadership

Over 5 years

Programming Languages TypeScript | Python | CSS

Libraries React | Robot Operating System | protobuf | Qt

Platforms Embedded Linux

Tools Docker | NoSQL | node.js | Kubernetes

Competencies People Management | Agile | REST / APIs | Microservice Architectures

Over 2 years

Programming Languages Rust | LaTeX

Libraries gRPC | CUDA | OpenGL

Competencies Agentic Workflows | Diffusion Models | GenAI